

C++ Programming

1. What is the main difference between C and C++?

- A) C++ is a procedural language; C is an object-oriented language.
- B) C++ supports Object-Oriented Programming (OOP); C does not.
- C) C++ only uses pointers; C uses references.
- D) C++ is interpreted; C is compiled.

Answer: B) C++ supports Object-Oriented Programming (OOP); C does not.

2. Which header file is typically used for input and output operations in C++?

- A) `<stdio.h>`
- B) `<iostream>`
- C) `<stream.h>`
- D) `<stdlib.h>`

Answer: B) `<iostream>`

3. Which operator is used for dynamic memory allocation in C++?

- A) `malloc()`
- B) `create`
- C) `new`
- D) `allocate`

Answer: C) `new`

4. Which keyword is used to deallocate memory allocated by the `new` operator?

- A) `free()`
- B) `delete`
- C) `remove`
- D) `release`

Answer: B) `delete`

5. What is a class in C++?

- A) A memory block to store variables.
- B) A blueprint for creating objects.
- C) A function definition.
- D) A collection of global constants.

Answer: B) A blueprint for creating objects.

6. Which concept bundles data and methods that operate on the data into a single unit?

- A) Inheritance
- B) Polymorphism
- C) Encapsulation
- D) Abstraction

Answer: C) Encapsulation

7. Which access specifier allows members to be accessed only by methods within the same class?

- A) `public`
- B) `protected`
- C) `private`
- D) `internal`

Answer: C) `private`

8. What is the purpose of a constructor in a C++ class?

- A) To destroy the object when it goes out of scope.
- B) To initialize the object's data members.
- C) To return a value from the class.
- D) To define the object's public methods.

Answer: B) To initialize the object's data members.

9. A constructor that takes no arguments is called the:

- A) Parameterized constructor
- B) Copy constructor

- C) Default constructor
- D) Initializer constructor

Answer: C) Default constructor

10. Which keyword is used to implement inheritance in C++?

- A) `implements`
- B) `inherits`
- C) `:` (colon)
- D) `extends`

Answer: C) `:` (colon)

11. What is the process of defining two or more functions with the same name but different parameter lists?

- A) Function Overriding
- B) Function Overloading
- C) Dynamic Binding
- D) Method Hiding

Answer: B) Function Overloading

12. What is a virtual function?

- A) A function that cannot be overloaded.
- B) A function whose address is known at compile time.
- C) A function declared with the `virtual` keyword, enabling runtime polymorphism.
- D) A function that always returns void.

Answer: C) A function declared with the `virtual` keyword, enabling runtime polymorphism.

13. What is a pure virtual function used for?

- A) To make a function non-overrideable.
- B) To allow the function to return multiple types.
- C) To make a class abstract.
- D) To ensure the function is only called once.

Answer: C) To make a class abstract.

14. Which feature allows an object of a derived class to be treated as an object of its base class?

- A) Encapsulation
- B) Static binding
- C) Run-time Polymorphism
- D) Inheritance of private members

Answer: C) Run-time Polymorphism

15. What is the type of a variable declared as `int &r = x;`?

- A) Pointer
- B) Reference
- C) Array
- D) Constant

Answer: B) Reference

16. Which operator is typically overloaded to work with `cout`?

- A) `>>` (Extraction operator)
- B) `<<` (Insertion operator)
- C) `+` (Addition operator)
- D) `&` (Address operator)

Answer: B) `<<` (Insertion operator)

17. What is the use of the scope resolution operator (`::`)?

- A) To access members of a structure.
- B) To define the scope of a loop.
- C) To access a class's static members or define a method outside the class body.
- D) To perform pointer arithmetic.

Answer: C) To access a class's static members or define a method outside the class body.

18. The Standard Template Library (STL) in C++ primarily contains:

- A) Only algorithms
- B) Containers, Algorithms, and Iterators
- C) Only I/O functions
- D) Operating System functions

Answer: B) Containers, Algorithms, and Iterators

19. Which STL container provides contiguous memory storage like an array but can dynamically resize?

- A) ``std::list``
- B) ``std::map``
- C) ``std::vector``
- D) ``std::deque``

Answer: C) ``std::vector``

20. Which STL container stores elements in a sorted order based on a key and typically uses a balanced binary search tree?

- A) ``std::vector``
- B) ``std::list``
- C) ``std::map``
- D) ``std::stack``

Answer: C) ``std::map``

21. What is the term for a generic function or class that operates on different data types?

- A) Overloaded
- B) Dynamic
- C) Templated
- D) Abstract

Answer: C) Templated

22. Which keyword is used to declare a template?

- A) ``typedef``
- B) ``generic``

C) ``template``

D) ``type``

Answer: C) ``template``

23. The mechanism to handle runtime errors in C++ is called:

A) Assertions

B) Preprocessor directives

C) Exception Handling

D) Debugging

Answer: C) Exception Handling

24. Which keyword is used to signal that an error has occurred?

A) ``try``

B) ``catch``

C) ``throw``

D) ``handle``

Answer: C) ``throw``

25. What is the role of the ``catch(...)`` block?

A) Catches only specific integer exceptions.

B) Catches only specific string exceptions.

C) Catches any type of exception (the universal handler).

D) Catches only I/O exceptions.

Answer: C) Catches any type of exception (the universal handler).

26. Which keyword is used to define a block of code where exceptions might occur?

A) ``catch``

B) ``throw``

C) ``try``

D) ``finally``

Answer: C) ``try``

27. What is the difference between a pointer and a reference in C++?

- A) A reference can be null, a pointer cannot.
- B) A pointer must be initialized; a reference does not need to be.
- C) A reference cannot be reseated to refer to another variable after initialization; a pointer can.
- D) References consume more memory than pointers.

Answer: C) A reference cannot be reseated to refer to another variable after initialization; a pointer can.

28. The use of `std::cout` and `std::cin` over `printf()` and `scanf()` provides the benefit of:

- A) Faster execution speed.
- B) Type-safety and extensibility through operator overloading.
- C) Smaller program size.
- D) Simpler syntax for all data types.

Answer: B) Type-safety and extensibility through operator overloading.

29. What is a destructor?

- A) A special member function called when an object is created.
- B) A special member function called when an object is destroyed or goes out of scope.
- C) Any function that returns void.
- D) A function that cannot be inherited.

Answer: B) A special member function called when an object is destroyed or goes out of scope.

30. How is a destructor named?

- A) Same as the class name.
- B) Class name preceded by a `#` sign.
- C) Class name preceded by a `~` (tilde) sign.
- D) `delete_` followed by the class name.

Answer: C) Class name preceded by a `~` (tilde) sign.

31. If a base class destructor is not virtual, what issue can occur during dynamic allocation?

- A) Compile-time error.
- B) Memory leak (derived class destructor is not called).
- C) Incorrect object creation.
- D) The base class destructor is called twice.

Answer: B) Memory leak (derived class destructor is not called).

32. The `this` pointer in a member function points to:

- A) The class definition.
- B) The current object for which the function is called.
- C) The nearest base class object.
- D) The memory address of the first static member.

Answer: B) The current object for which the function is called.

33. Which type of polymorphism is achieved through function overloading?

- A) Compile-time polymorphism (Static binding)
- B) Run-time polymorphism (Dynamic binding)
- C) Behavioral polymorphism
- D) Template polymorphism

Answer: A) Compile-time polymorphism (Static binding)

34. A Friend Function of a class is:

- A) Always a member of another class.
- B) A non-member function that has access to the class's `private` and `protected` members.
- C) A member function that can only access `public` members.
- D) An inline function that performs I/O.

Answer: B) A non-member function that has access to the class's `private` and `protected` members.

35. The `protected` access specifier allows access by:

- A) Only the class members.
- B) Class members and members of derived classes.

C) Class members and any external function.

D) Class members, derived class members, and friend functions.

Answer: D) Class members, derived class members, and friend functions.

36. What is the primary advantage of using a namespace?

A) To reduce program execution time.

B) To prevent name collisions between different libraries or components.

C) To define global variables.

D) To enable function overriding.

Answer: B) To prevent name collisions between different libraries or components.

37. Which statement is used to bring a specific item from a namespace into the current scope?

A) ``include``

B) ``using namespace``

C) ``using``

D) ``alias``

Answer: C) ``using``

38. What is the problem that can arise from Multiple Inheritance where a class inherits from two base classes that share a common ancestor?

A) Circular Dependency

B) Diamond Problem

C) Initialization Error

D) Dangling Pointer

Answer: B) Diamond Problem

39. What is an Iterator in STL?

A) A type of container that stores sequential elements.

B) An object that acts like a pointer to elements in a container.

C) A generic algorithm for sorting.

D) A function that manages memory.

Answer: B) An object that acts like a pointer to elements in a container.

40. Which smart pointer automatically deletes the memory when the pointer goes out of scope and does not allow copying?

- A) `std::shared_ptr`
- B) `std::weak_ptr`
- C) `std::unique_ptr`
- D) `std::auto_ptr` (deprecated)

Answer: C) `std::unique_ptr`

41. Which operator is used to access the members of an object using a pointer?

- A) `.` (dot operator)
- B) `::` (scope resolution)
- C) `->` (arrow operator)
- D) `*` (dereference operator)

Answer: C) `->` (arrow operator)

42. What happens if a function is declared as inline?

- A) The compiler must treat it as a virtual function.
- B) The function code is inserted directly at the point of call to reduce overhead.
- C) It is only visible within the current source file.
- D) It can only be called once.

Answer: B) The function code is inserted directly at the point of call to reduce overhead.

43. Which keyword in C++ is used to prevent a class from being inherited?

- A) `final` (C++11/14)
- B) `sealed`
- C) `private` (inheritance)
- D) `const`

Answer: A) `final` (C++11/14)

44. The `std::endl` manipulator used with `cout` does what two things?

- A) Flushes the buffer and inserts a space.
- B) Inserts a newline character and flushes the buffer.
- C) Inserts a newline character and terminates the program.
- D) Inserts a space and flushes the buffer.

Answer: B) Inserts a newline character and flushes the buffer.

45. The `std::string` class is part of which header file?

- A) `<cstring>`
- B) `<string.h>`
- C) `<iostream>`
- D) `<string>`

Answer: D) `<string>`

46. What does the `static` keyword mean when used with a class member variable?

- A) The variable is stored on the stack.
- B) The variable retains its value between function calls.
- C) The variable is shared by all objects of the class.
- D) The variable cannot be modified.

Answer: C) The variable is shared by all objects of the class.

47. What is an initializer list in C++ used for?

- A) To initialize static members.
- B) To initialize an array of objects.
- C) To initialize data members in a constructor, especially for const or reference members.
- D) To define the order of function calls.

Answer: C) To initialize data members in a constructor, especially for const or reference members.

48. Which stream class is used to handle file output operations?

- A) `ifstream`
- B) `ostream`

C) ``fstream``

D) ``ofstream``

Answer: D) ``ofstream``

49. What is the C++-specific syntax for comments that span a single line?

A) ``//``

B) ``##``

C) ``--``

D) ``/* ... */``

Answer: A) ``//``

50. The Rule of Three/Five in C++ relates to which set of functions?

A) Operator overloads for ``+``, ``-``, and ``*``.

B) Virtual, Inline, and Friend functions.

C) Constructor, Destructor, Copy Constructor, Copy Assignment Operator, and Move Operations.

D) Public, Protected, and Private access specifiers.

Answer: C) Constructor, Destructor, Copy Constructor, Copy Assignment Operator, and Move Operations.

51. What is the term for a C++ function that is declared as a member of a class but has no implementation, and the class cannot be instantiated?

A) Abstract function

B) Pure virtual function

C) Static member function

D) Destructor

Answer: B) Pure virtual function

52. The ``auto`` keyword (C++11) is used for:

A) Declaring a variable with automatic storage duration.

B) Declaring a variable whose type is automatically deduced from its initializer.

C) Declaring a function with a default return type.

D) Declaring a global variable.

Answer: B) Declaring a variable whose type is automatically deduced from its initializer.

53. What is a Lambda Expression (C++11)?

- A) An anonymous inline function object that can be passed as an argument.
- B) A new type of loop construct.
- C) A complex template specialization.
- D) A way to define a static class member.

Answer: A) An anonymous inline function object that can be passed as an argument.

54. The capture clause `[&]` in a lambda expression indicates:

- A) Capture by value.
- B) Capture by reference.
- C) Capture all local variables by value.
- D) Capture all local variables by reference.

Answer: D) Capture all local variables by reference.

55. Which header file contains the definition for `std::vector`?

- A) ``
- B) `
`
- C) ``
- D) ``

Answer: C) ``

56. Which STL algorithm is used to sort a container in C++?

- A) `qsort()`
- B) `std::sort()`
- C) `std::quicksort()`
- D) `std::arrange()`

Answer: B) `std::sort()`

57. In the context of inheritance, the `protected` members of the base class become what in a derived class using public inheritance?

- A) Private
- B) Protected
- C) Public
- D) Unavailable

Answer: B) Protected

58. What is the result of applying the prefix increment operator (`++x`)?

- A) The value is incremented, and the original value is returned.
- B) The value is incremented, and the new value is returned.
- C) The value is not incremented, and the original value is returned.
- D) It is only applicable to pointers.

Answer: B) The value is incremented, and the new value is returned.

59. When overloading the pre-increment operator (`++`), what is the signature of the non-member function?

- A) `operator++ (T& obj)`
- B) `operator++ (T obj)`
- C) `operator++ (int)`
- D) `operator++ ()`

Answer: D) `operator++ ()`

60. What is a Copy Constructor?

- A) A constructor that initializes an object from another object of the same class.
- B) A constructor that creates a shallow copy.
- C) A constructor that only takes one integer argument.
- D) A constructor used only in multiple inheritance.

Answer: A) A constructor that initializes an object from another object of the same class.

61. What is the primary benefit of using `std::shared\_ptr`?

- A) Guaranteed single ownership of the resource.

- B) Prevents memory leaks by implementing reference counting.
- C) Faster than raw pointers.
- D) Stores the resource on the stack.

Answer: B) Prevents memory leaks by implementing reference counting.

62. Which of the following is an example of an unqualified name?

- A) ``std::cout``
- B) ``MyClass::member``
- C) ``variable_name``
- D) ``::global_var``

Answer: C) ``variable_name``

63. What does the ``explicit`` keyword prevent a constructor from being used for?

- A) Default construction.
- B) Parameterized construction.
- C) Implicit type conversion.
- D) Copy construction.

Answer: C) Implicit type conversion.

64. What is the function of ``const_cast``?

- A) To convert between a pointer to a base class and a pointer to a derived class.
- B) To remove or add the ``const`` or ``volatile`` qualifiers from a pointer or reference.
- C) To convert between unrelated types.
- D) To cast between fundamental types (e.g., ``int`` to ``float``).

Answer: B) To remove or add the ``const`` or ``volatile`` qualifiers from a pointer or reference.

65. Which loop construct is new in C++11 and designed to easily iterate over elements in a container?

- A) ``while`` loop
- B) Range-based ``for`` loop
- C) ``do-while`` loop

D) ``goto` loop`

Answer: B) Range-based ``for` loop`

66. What is the size of an empty class object in C++?

A) 0 bytes

B) 1 byte

C) 4 bytes

D) Compiler dependent, typically 1 byte for unique addressability.

Answer: B) 1 byte

67. Which function from the `<algorithm>` header is used to find an element in a range?

A) ``search()``

B) ``find()``

C) ``locate()``

D) ``position()``

Answer: B) ``find()``

68. What is the key difference between ``std::list`` and ``std::vector``?

A) ``std::list`` is faster for random access.

B) ``std::vector`` provides constant time ($O(1)$) insertion/deletion at any position.

C) ``std::list`` is a doubly linked list; ``std::vector`` is a dynamic array.

D) ``std::list`` uses contiguous memory; ``std::vector`` does not.

Answer: C) ``std::list`` is a doubly linked list; ``std::vector`` is a dynamic array.

69. What is the process of resolving virtual function calls at run-time?

A) Static Binding

B) Early Binding

C) Late Binding (Dynamic Dispatch)

D) Inline expansion

Answer: C) Late Binding (Dynamic Dispatch)

70. The copy-and-swap idiom is primarily used to implement which feature correctly?

- A) Inheritance
- B) Function overloading
- C) Strong exception safety for assignment operators
- D) Recursive functions

Answer: C) Strong exception safety for assignment operators

71. What does RAII (Resource Acquisition Is Initialization) principle advocate?

- A) Resources should be acquired in the main function.
- B) Resources should be released as soon as possible.
- C) Resources should be managed by object lifetimes (e.g., in constructors and destructors).
- D) Resources should only be allocated using `malloc()`.

Answer: C) Resources should be managed by object lifetimes (e.g., in constructors and destructors).

72. What are Rvalue references (C++11) primarily used to enable?

- A) Copying objects
- B) Move semantics
- C) Run-time type information
- D) Thread safety

Answer: B) Move semantics

73. The `std::move()` function in C++ is used for:

- A) Physically relocating an object in memory.
- B) Casting an lvalue into an rvalue reference, enabling resource theft (moving resources).
- C) Swapping two objects.
- D) Deleting an object.

Answer: B) Casting an lvalue into an rvalue reference, enabling resource theft (moving resources).

74. What is the main purpose of `std::weak_ptr`?

- A) To manage resources with strong ownership.

- B) To break circular references between `std::shared_ptr` objects.
- C) To hold the address of a local variable.
- D) To replace raw pointers completely.

Answer: B) To break circular references between `std::shared_ptr` objects.

75. What does the `using namespace std;` statement do?

- A) Makes all standard library functions inline.
- B) Makes all standard library names available without the `std::` prefix.
- C) Includes all standard header files.
- D) Sets the default memory allocator to the standard one.

Answer: B) Makes all standard library names available without the `std::` prefix.

76. The `static_cast` operator is used for:

- A) Converting between unrelated types.
- B) Casting pointers to polymorphic types at run-time.
- C) Casting between related types in an inheritance hierarchy or performing safe conversions (e.g., `int` to `float`).
- D) Casting away `const`ness.

Answer: C) Casting between related types in an inheritance hierarchy or performing safe conversions (e.g., `int` to `float`).

77. Which C++ operator is used to determine the exact type of an object at runtime (Run-Time Type Information - RTTI)?

- A) `sizeof`
- B) `typeid`
- C) `typeof`
- D) `dynamic_cast`

Answer: B) `typeid`

78. What does `dynamic_cast` return if the cast from a base-class pointer/reference to a derived-class pointer/reference fails?

- A) A compile-time error.

B) An exception.

C) A null pointer (for pointer cast) or throws an exception (for reference cast).

D) The address of the base object.

Answer: C) A null pointer (for pointer cast) or throws an exception (for reference cast).

79. What is a functor (function object) in C++?

A) A class that overrides the function call operator (`operator()`).

B) A function that returns another function.

C) A static function that takes a pointer.

D) A function defined within a lambda expression.

Answer: A) A class that overrides the function call operator (`operator()`).

80. What is the purpose of the `mutable` keyword?

A) To make a member variable accessible only by friend functions.

B) To allow a data member of a `const` object to be modified.

C) To allow a function to be declared as `virtual`.

D) To allow a variable to be stored in a CPU register.

Answer: B) To allow a data member of a `const` object to be modified.

81. Which C++ I/O manipulator is used to clear the format flags on a stream?

A) `setf`

B) `unsetf`

C) `resetiosflags`

D) `flush`

Answer: C) `resetiosflags`

82. What is the output of the following C++ code?

<!-- end list -->

```cpp

```
int i = 10;
int *p = &i;
std::cout << *p++;
...
```

A) 10 (and then `p` is incremented)

B) 11

C) 10 (and `i` is incremented)

D) Compiler error

Answer: A) 10 (and then `p` is incremented)

83. What is the output of the following C++ code snippet?

<!-- end list -->

```
```cpp
int a = 5;
int &r = a;
r++;
std::cout << a;
...
```

A) 5

B) 6

C) 7

D) Error

Answer: B) 6

84. The process of defining a derived class method that has the exact same name, return type, and parameters as a base class virtual method is called:

A) Overloading

- B) Hiding
- C) Overriding
- D) Polymorphing

Answer: C) Overriding

85. Which type of header file is generally used to include user-defined headers?

- A) `<filename>`
- B) `"filename"`
- C) `[filename]`
- D) `//filename`

Answer: B) `"filename"`

86. The Diamond Problem in multiple inheritance can be resolved in C++ using which type of inheritance?

- A) Multilevel Inheritance
- B) Public Inheritance
- C) Virtual Inheritance
- D) Private Inheritance

Answer: C) Virtual Inheritance

87. Which container provides FIFO (First-In, First-Out) access to its elements?

- A) `std::stack`
- B) `std::vector`
- C) `std::queue`
- D) `std::priority_queue`

Answer: C) `std::queue`

88. Which container provides LIFO (Last-In, First-Out) access to its elements?

- A) `std::list`
- B) `std::deque`
- C) `std::set`

D) ``std::stack``

Answer: D) ``std::stack``

89. When a ``const`` member function is called, what does the ``this`` pointer become?

A) ``const``

B) ``volatile``

C) ``const ClassName*``

D) ``ClassName* const``

Answer: C) ``const ClassName*``

90. The ``extern "C"`` linkage specifier is used to:

A) Ensure a function is defined in a C++ file.

B) Tell the compiler to use C naming and calling conventions for the specified function.

C) Link to external libraries written in C++.

D) Define a C-style string.

Answer: B) Tell the compiler to use C naming and calling conventions for the specified function.

91. What is the C++-style way to check if a file stream (``ifstream`` or ``ofstream``) is successfully opened?

A) Check its ``status()`` method.

B) Use the ``is_open()`` method or check the stream object directly in an ``if`` condition.

C) Compare the stream object to ``NULL``.

D) Check the global variable ``errno``.

Answer: B) Use the ``is_open()`` method or check the stream object directly in an ``if`` condition.

92. Which function is used to convert a C++ ``std::string`` object to a C-style string (``const char*``)?

A) ``c_str()``

B) ``to_c_string()``

C) ``convert()``

D) ``data()``

Answer: A) ``c_str()``

93. The concept of Aggregation in OOP represents:

- A) A "is-a" relationship.
- B) A "has-a" relationship, where parts can exist independently of the whole.
- C) A strong "is composed of" relationship.
- D) A one-to-one relationship.

Answer: B) A "has-a" relationship, where parts can exist independently of the whole.

94. What is the main characteristic of Composition?

- A) The parts can be shared between multiple wholes.
- B) The lifetime of the part is dependent on the lifetime of the whole.
- C) It is achieved through virtual functions.
- D) It always involves multiple inheritance.

Answer: B) The lifetime of the part is dependent on the lifetime of the whole.

95. In C++, which operator has the lowest precedence?

- A) Comma operator (`,`)
- B) Assignment operator (`=`)
- C) Logical OR (`||`)
- D) Conditional operator (`?:`)

Answer: A) Comma operator (`,`)

96. What is the purpose of the `default` keyword in a switch statement?

- A) To specify the first case to be executed.
- B) To handle values that are not explicitly matched by any `case` label.
- C) To define a default constructor.
- D) To set a variable to its default value.

Answer: B) To handle values that are not explicitly matched by any `case` label.

97. The concept of Data Hiding is achieved primarily through which C++ feature?

- A) Pointers

- B) Templates
- C) Access specifiers (`private` and `protected`)
- D) Namespaces

Answer: C) Access specifiers (`private` and `protected`)

98. Which keyword is used to explicitly declare a member function as deleted in C++11 to prevent its use (e.g., prevent copy construction)?

- A) `delete`
- B) `disallow`
- C) `= default`
- D) `= delete`

Answer: D) `= delete`

99. Which STL container is typically implemented as a hash table?

- A) `std::set`
- B) `std::unordered\_map`
- C) `std::map`
- D) `std::list`

Answer: B) `std::unordered\_map`

100. What is the output of the following C++ code?

<!-- end list -->

```
```cpp
int a = 10;
std::cout << (a++ * 2);
```
```

- A) 22
- B) 20

C) 21

D) 12

Answer: B) 20